STATEMENT OF WORK
4th Bundle Replacement
FCI WASECA - B&F PROJECT #26Z4AN6

A. INTRODUCTION:

1. The Federal Bureau of Prisons (FBOP), FCI Waseca, intends to solicit the purchase and delivery of one instantaneous water heater with thermostatic mixing valve, hot water holding tank.

2. Supply vendor must comply with all applicable Federal, State and Local building codes and regulations. This contract is located at the Federal Correctional Institution (FCI) Waseca, 1000 University Drive SW Waseca, MN 56093.

3. All equipment specified in this statement must be equal or better product. Should your company choose to submit an equal product, you must submit all technical specifications of the equal product for review and determination shall be issued with acceptance or non-acceptance. The government retains the right to determine technical acceptance for all submitted equal products. This solicitation is a supply only.

B. EXISTING CONDITIONS:

1. FCI Waseca currently has three existing instantaneous water heaters that supply hot water to the institution. New laundry equipment has been added, and the current system cannot keep up with the demand.

C. REQUIREMENTS:

1. Quantity Requirement:

    a. (1) one 50 GPM instantaneous steam/water heater with thermostatic mixing valve

    b. (1) one 275-gallon hot water storage tank

D. SPECIFICATIONS / PRODUCT FEATURES / CHARACTERISTICS:

1. Product Basis:

    a. Armstrong RT66540R Flo-Rite-Temp Steam to Water Heat Exchanger, or approved equal

    b. Armstrong 275-gallon hot water storage tank 40” x 78” insulated, or approved equal

2. Refer to attachment 1 for the instantaneous steam/water heater specifications.

3. Refer to attachment 2 for the hot water storage tank cut sheet.

E. QUALITY ASSURANCE / START UP WARRANTY (IF REQUIRED):

1. The Supply Vendor warrants that the equipment provided shall be consistent with current industry standards and the International Plumbing Code (IPC), Underwriters Laboratories (UL), and the National Standard Plumbing Code (NSPC).

2. The equipment shall be free from any defects in materials and workmanship for a period of at least one (1) year from the date of final acceptance. During this warranty period, the Supply Vendor shall, at no additional cost to

FBOP, correct any defects in the products that are identified. This warranty includes, but is not limited to, the following:

a. Materials and Equipment: All materials and equipment provided by the Supplier will be new, of good quality, and free from defects.

b. Compliance: All equipment will comply with the specifications and requirements set forth in this SOW and any related documents.

3. If any defects are discovered during the warranty period, the FBOP shall notify the Vendor in writing. The Vendor will respond promptly and take necessary actions to correct the defects within a reasonable timeframe

F. SUBMITTALS:

1. The Supply Vendor shall provide FBOP with all the technical documents of the equipment including operation manuals, product data sheets, warranty documents, and manufacturer's support agreement upon, or prior to purchase agreement.

2. The Vendor shall provide to the Contract Monitor (3) copies of the Material Safety Data Sheets (MSDS) on all materials and substances that may be used during the project in advance. In the event the material is not approved by the Safety Manager, it shall be the responsibility of the supply vendor to locate and provide an alternative product at no additional cost to the government.

G. DELIVERY:

1. Delivery hours for supply vendors are **8:00 AM – 2:00 PM**, Monday through Friday, excluding weekends and federal holidays. Any delivery required to be completed at any other time than noted above must be scheduled five (5) working days prior to the start of the work. Only a facilities department staff members is authorized to receive or sign for any deliveries made to the site.

H. ADDITIONAL CONSIDERATIONS:

1. Delivery personnel will NOT be allowed to take pictures of the Institution, especially where inmates are present during delivery at FCI Waseca.

2. Delivery personnel will be permitted to wear jeans in the Institution; however, the Institution will specify which colors of clothing shall not be allowed. The clothes shall be proper and suitable for the services that are being provided.

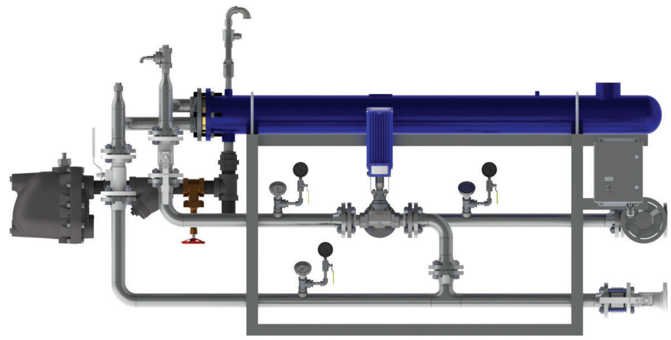
3. Smoking including e-cigarettes and vaping devices is prohibited on Federal property. There are some exceptions and additional restrictions for Institutions that have designated smoking areas.

Attachment 1

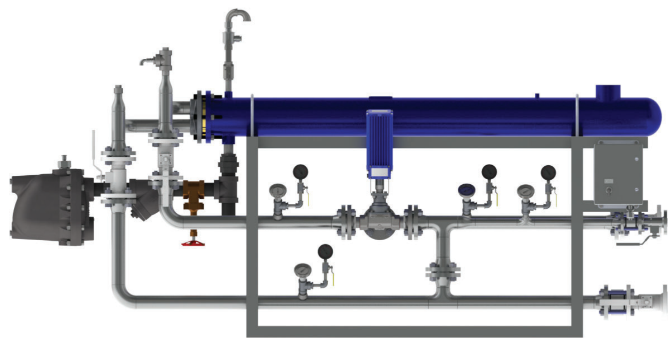
Hot Water Generation - Steam to Water

ReadiTemp™ Shell and Tube Heat Exchanger Model RT66540G & RT66540R

A complete pre-piped instantaneous steam/water shell and tube heater with Emech® Digital Control Valve (E40W).
RT66540G is designed for non-recirculated systems.
RT66540R is designed for recirculated systems.
The E40W utilizes ceramic shear action disc technology to provide high pressure differential capability and log life integrity. Fitted with the electronic actuator and integrated temperature sensor, the system delivers high performance stand-alone closed-loop temperature control. Even with sudden changes of inlet pressure and temperature to the valve, the actuator controller aggressively minimizes outlet temperature variations, making the system ideal for use in process applications.



RT66540G



RT66540R

Operational Specifications

- Operating temperature range: -13°F to 257°F (-25°C to 125°C)
- ± 1°F (± 0.5°C) over a 32°F - 212°F (0°C - 100°C) control range
- Capable of positioning from 0% to 100% of either inlet temperature
- 90° stroke time as low as 1.5 seconds for fast control action
- 100% duty cycle rated actuator
- Capable of control blending to within 1°F of either hot or cold inlet temperature

Technical Specifications

- Maximum domestic pressure: 145 PSI
- Maximum steam pressure: 15 PSIG
- Minimum flow rate: 7 GPM
- Design conditions: At 80°F temperature rise 85 GPM using 15 PSIG steam at 3,601 lbs/hr; at 100°F temperature rise 63 GPM using 15 PSIG steam at 3,338 lbs/hr; at 120°F temperature rise 48 GPM using 15 PSIG steam at 3,057 lbs/hr
- Single wall heat exchanger tubes shall be straight 5/8” x .065” OD Admiralty Brass expanded into Naval Brass tube sheets with a free-floating end cover
- BUBBLE TIGHT zero seat leakage shut-off (exceeds ASME B16.104 and FCI 70.2, Classes V and VI standards)
- Design verification to ASME B16.34
- Power: 120VAC 1A
- Failsafe position feedback (non-contact absolute encoder)
- Keypad: 4 membrane switches with “dual touch” safety features
- Display: 3.5 digit LCD display with back light
- Gearbox: planetary, lifetime lubrication, low backlash

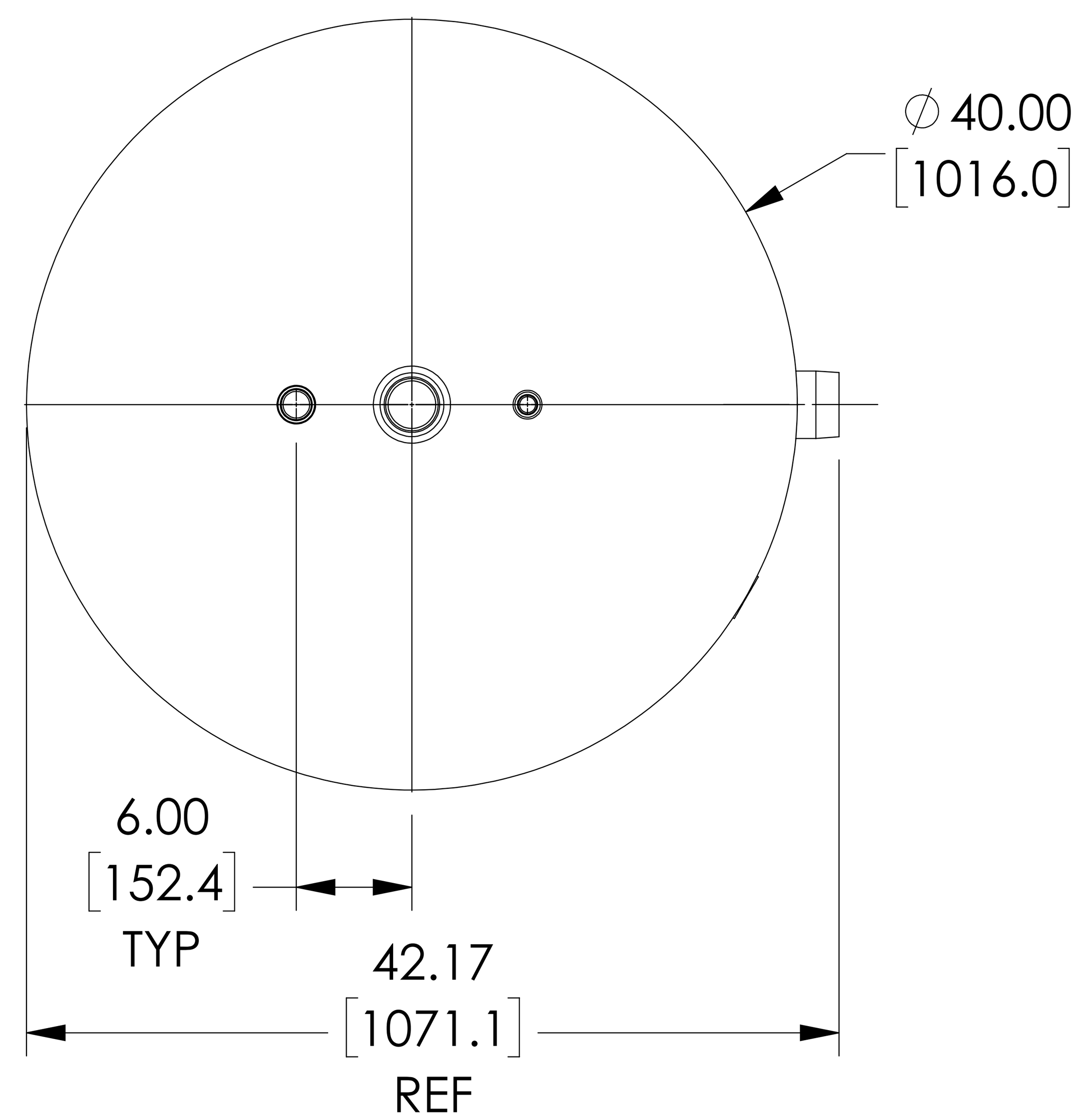
Standard Materials of Construction

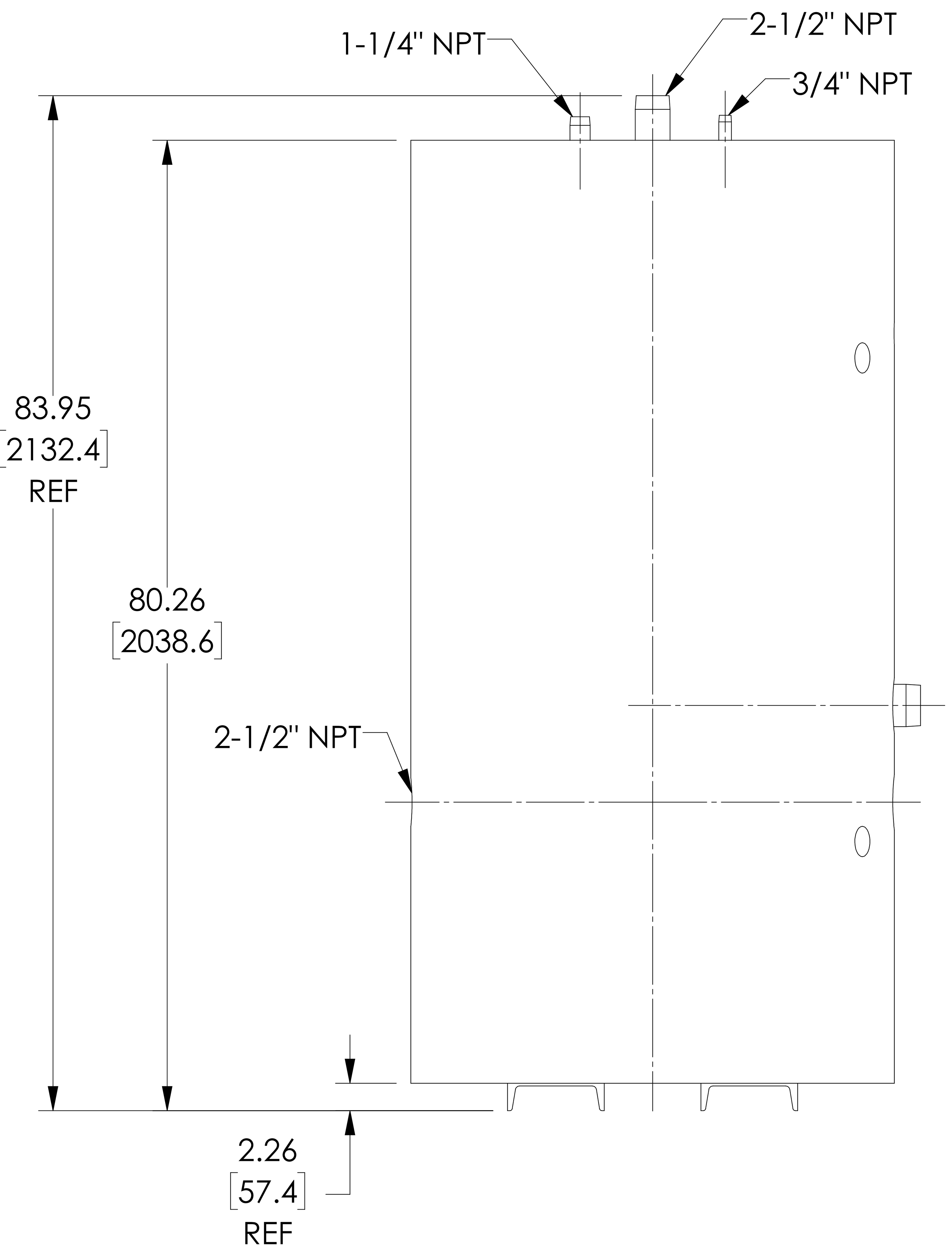
- Water Pipe: 316 Stainless Steel
- Fittings: 316 Stainless Steel
- Single wall tube bundle: Admiralty Brass
- Heat Exchanger Shell: Carbon Steel
- Steam Trap Model 20JD8 F&T: Ductile Iron
- Air Vent Model TS2: Brass
- Pressure Relief Valve: Stainless Steel
- Digital Control Valve: 316 Stainless Steel
- Digital Control Valve Discs: Ceramic, Durable Corrosion Resistant
- Frame: Fabricated Stainless Steel

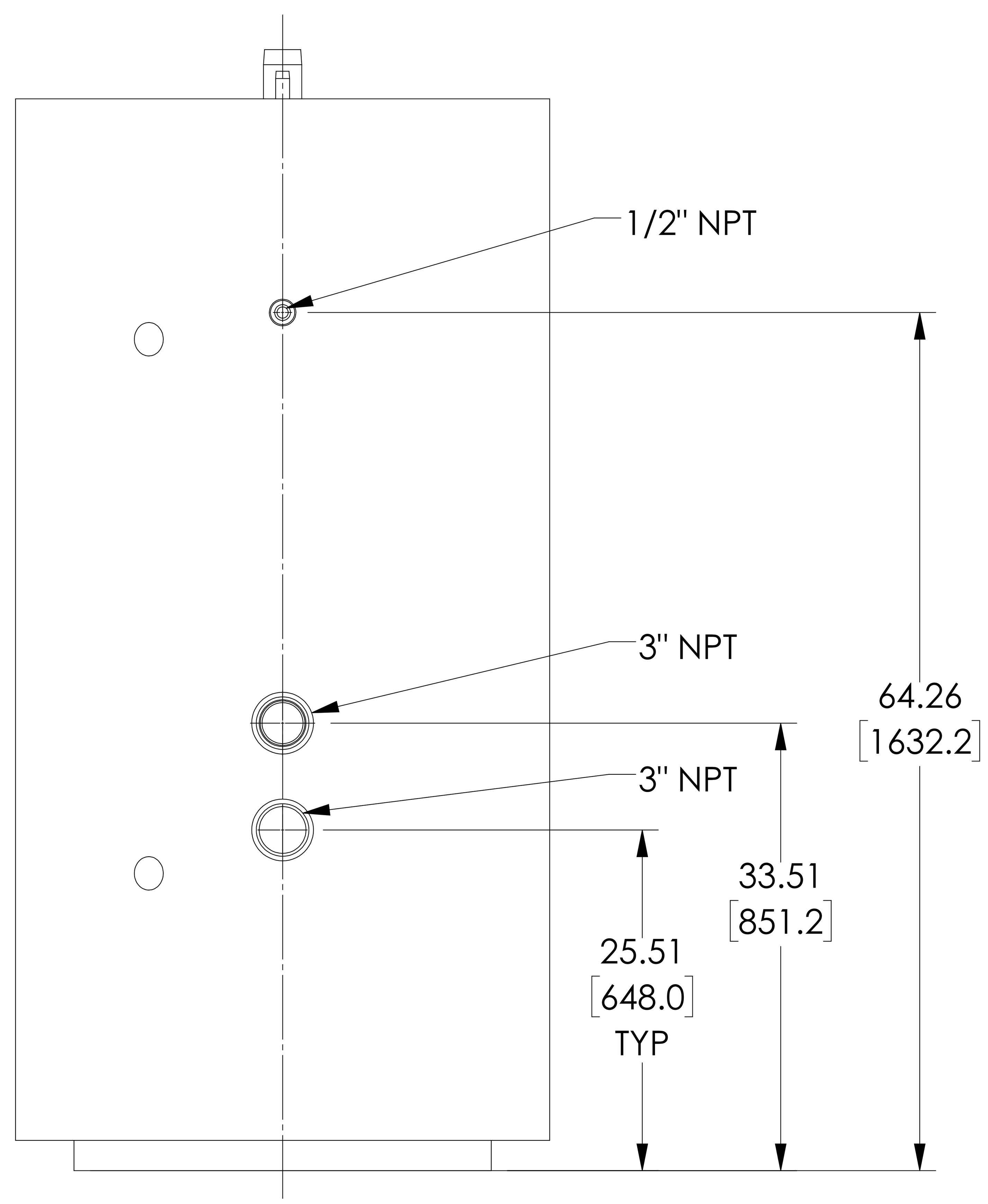
Connectivity

- Analog (4-20mA) input and output control signals for interfacing with SCADA control
- External RS232 connection for actuator configuration
- Stand-alone closed loop temperature control, or remote analog (4-20mA) control options
- Extra digital input for interfacing ancillary devices (e.g., flow switch, level switch)

Armstrong Hot Water Group, 221 Armstrong Blvd., Three Rivers, MI 49093 – USA Phone: 269-279-3602, Fax: 269-279-3130
armstronginternational.com







Attachment 2

NOTES:
1. WORKING PRESSURE: 125PSI [8.6 BAR]
2. THREADS PER B1.20.1

DO NOT SCALE DRAWING				
TOLERANCES UNLESS OTHERWISE SPECIFIED				
DIMENSIONING ENGLISH [mm]				
FRACTIONAL ± 1/64				
ANGULAR: ± 2				
			IN.	MM
DECIMAL	.XXXX	±	.0005	-----
	.XXX	±	.005	.010
	.XX	±	.015	.10
	.X		----------	.3

Armstrong®

	NAME	DATE
DRAWN	Kyle Bradford	12/19/2019
RELEASED		2/14/2020

ARMSTRONG INTERNATIONAL

TANK STO 275 GAL 40 X 78 INSUL JKT

MATERIAL			SHEET 1 OF 1
CN70364	REV A	DWG.	END124009